Deep Learning-based mmWave MIMO Channel Estimation using sub-6 GHz Channel Information: CNN and UNet Approaches

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Technische Universität Wien

Institute of Telecommunications

Introduction

- increasing demand for high data rate transmission

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- sub-6 GHz band cannot meet these requirements

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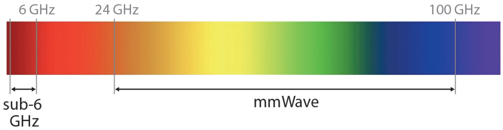
- sub-6 GHz band cannot meet these requirements
- mmWave bands allow for significantly higher data rates



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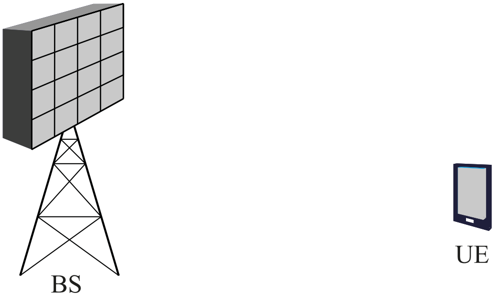
- mmWave systems utilize antenna arrays



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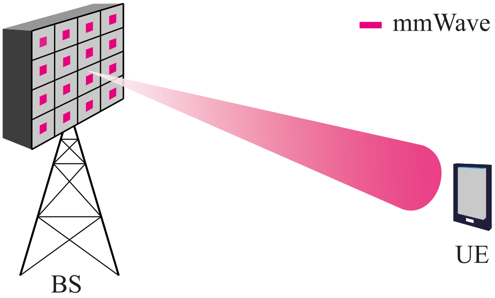
  - directional beamforming



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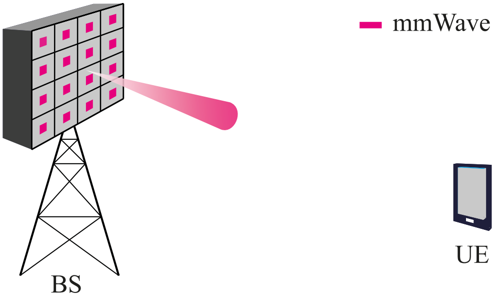
  - low pre-beamforming SNR



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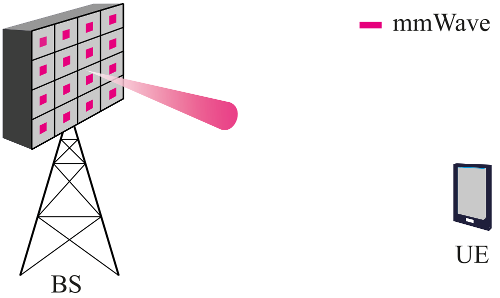

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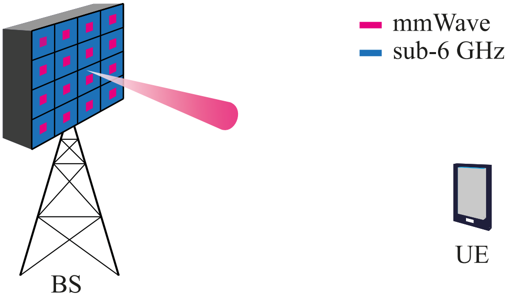

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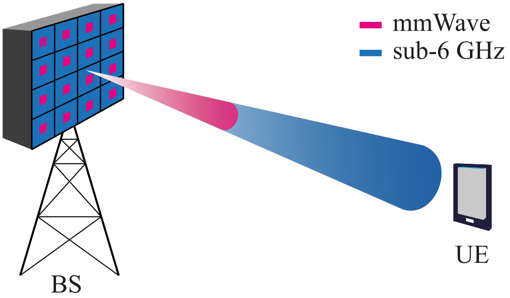
- mmWave deployment in conjunction with sub-6 GHz systems
  - use out-of-band channel information improve system throughput in the mmWave band [1]



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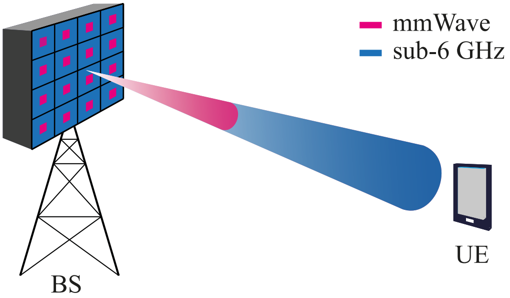
  - use out-of-band channel information improve system throughput in the mmWave band [1]

- many relationships between in-band and out-of-band channel remain unexplored



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Contribution

- **novel deep learning-based approach to estimate channel for mmWave MIMO communication systems**
  - use machine learning to combine conventional in-band and out-of-band aided channel estimates

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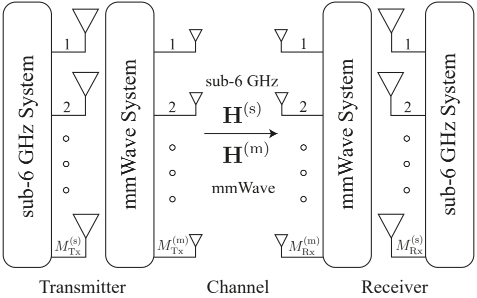
- **novel deep learning-based approach to estimate channel for mmWave MIMO communication systems**
  - use machine learning to combine conventional in-band and out-of-band aided channel estimates
- **evaluate proposed channel estimation methods through simulations**
  - achievable spectral efficiency (SE)

Contents

System Model

- multi-band MIMO OFDM system
  - co-located and aligned antenna arrays



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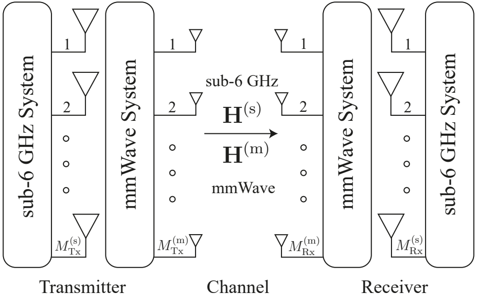

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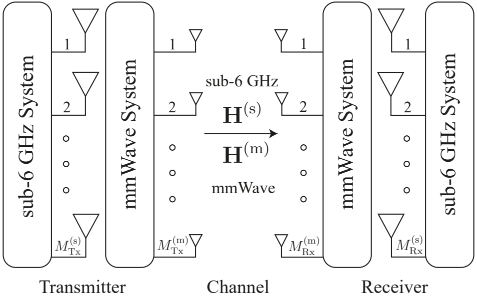

System Model

- multi-band MIMO OFDM system
  - co-located and aligned antenna arrays
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  - Ricean channel model

$$\mathbf{H}^{(\mathrm{s})} = \sqrt{\frac{K^{(\mathrm{s})}}{1+K^{(\mathrm{s})}}}\underbrace{\mathbf{H}_{\mathrm{fs}}^{(\mathrm{s})}}_{\text{free-space}} + \sqrt{\frac{1}{1+K^{(\mathrm{s})}}}\underbrace{\mathbf{H}_{\mathrm{rp}}^{(\mathrm{s})}}_{\text{Rayleigh part}}$$

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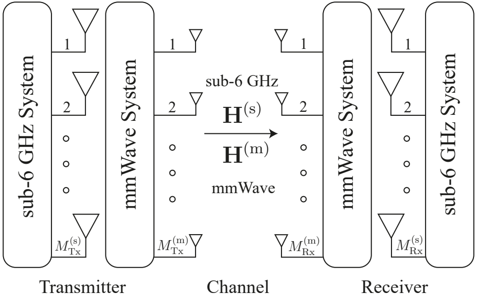

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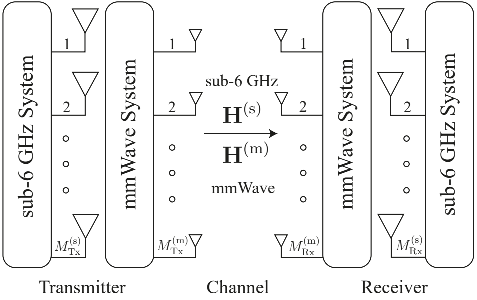

System Model

- multi-band MIMO OFDM system
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- link establishment
  - 1st training phase $\Longrightarrow$ $\widetilde{\mathbf{H}}^{(\mathrm{s})}$ and $\widetilde{\mathbf{H}}^{(\mathrm{m})}$

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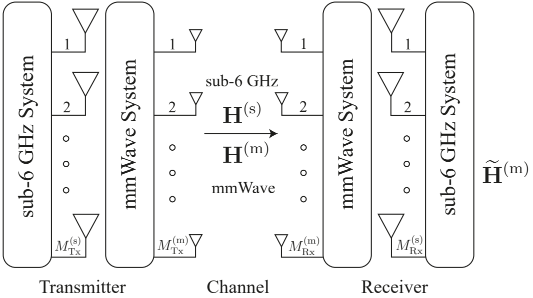

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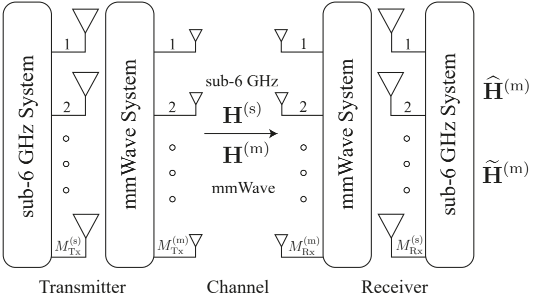

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  - resulting channel estimate: $\overline{\mathbf{H}}^{(\mathrm{m})}$

$$\mathbf{H}^{(\mathrm{s})} = \sqrt{\frac{K^{(\mathrm{s})}}{1+K^{(\mathrm{s})}}}\,\underbrace{\mathbf{H}^{(\mathrm{s})}_{\mathrm{fs}}}_{\text{free−space}} + \sqrt{\frac{1}{1+K^{(\mathrm{s})}}}\,\underbrace{\mathbf{H}^{(\mathrm{s})}_{\mathrm{rp}}}_{\text{Rayleigh part}}$$

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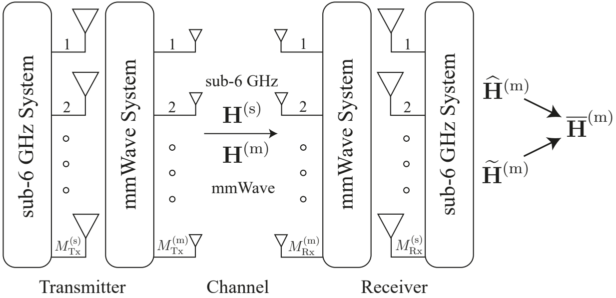

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  - resulting channel estimate: $\overline{\mathbf{H}}^{(\mathrm{m})}$
  - data transmission

$$\mathbf{y} = \overline{\mathbf{Q}}^{\mathrm{H}} \mathbf{H}^{(\mathrm{m})} \overline{\mathbf{F}} \mathbf{x} + \overline{\mathbf{Q}}^{\mathrm{H}} \mathbf{w}$$

receive symbol vector $\mathbf{y} \in \mathbb{C}^{\ell_{\max} \times 1}$
transmit symbol vector $\mathbf{x} \in \mathbb{C}^{\ell_{\max} \times 1}$
precoding matrix $\overline{\mathbf{F}} \in \mathbb{C}^{M_{\mathrm{Tx}} \times \ell_{\max}}$
combining matrix $\overline{\mathbf{Q}} \in \mathbb{C}^{M_{\mathrm{Rx}} \times \ell_{\max}}$
additive Gaussian noise $\mathbf{w} \in \mathbb{C}^{M_{\mathrm{Rx}} \times 1}$

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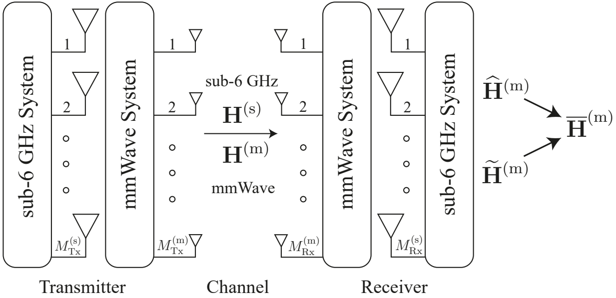

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Out-of-Band Aided Channel Estimation - Maximal Ratio Combining (MRC)

- gives an optimum performance for each combination of K-factor and noise variance [1]

$$\overline{\mathbf{H}}^{(\mathrm{m})}[n] = \widehat{w}\, \widehat{\mathbf{H}}^{(\mathrm{m})}[n] + (1-\widehat{w})\widetilde{\mathbf{H}}^{(\mathrm{m})}[n]$$

$$\widehat{w} \approx \frac{M_{\mathrm{Tx}}^{(\mathrm{m})} M_{\mathrm{Rx}}^{(\mathrm{m})} \sigma^2_{w^{(\mathrm{m})}}}{\frac{M_{\mathrm{Tx}}^{(\mathrm{m})} M_{\mathrm{Rx}}^{(\mathrm{m})}}{1+\widetilde{K}^{(\mathrm{s})}} + \left(1 + M_{\mathrm{Tx}}^{(\mathrm{m})} M_{\mathrm{Rx}}^{(\mathrm{m})}\right) \sigma^2_{w^{(\mathrm{m})}}}$$

estimated sub-6 GHz K-factor $\widetilde{K}^{(\mathrm{s})}$
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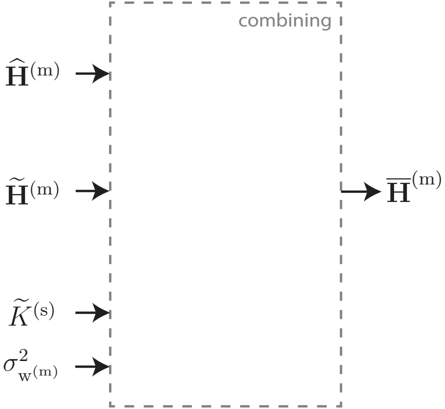
mmWave SNR $\gamma^{(\mathrm{m})}$



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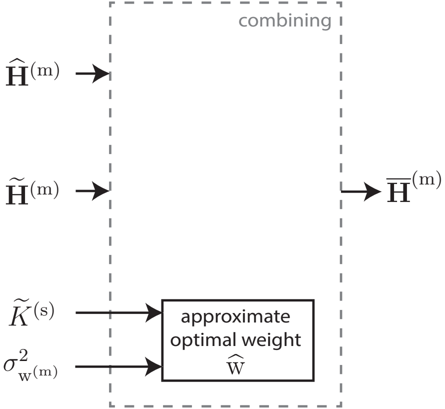

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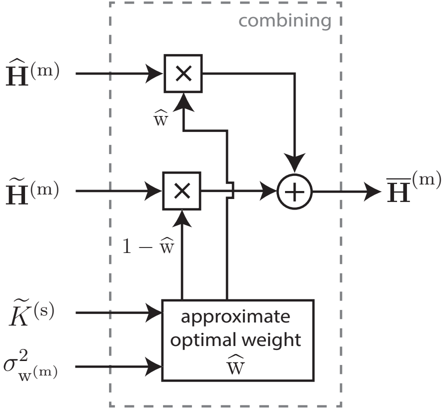

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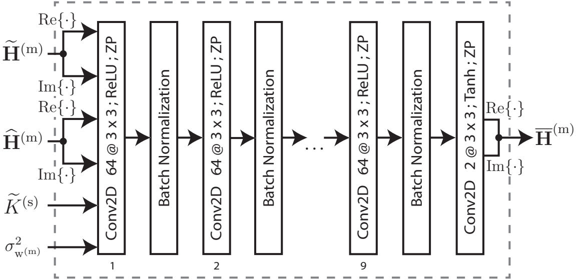
- input-output mapping relationship with the parameter set of Φ_{C}

$$\overline{\mathbf{H}}^{(\mathrm{m})}[n] = f_{\Phi_{\mathrm{C}}}\left(\widetilde{\mathbf{H}}^{(\mathrm{m})}[n], \widehat{\mathbf{H}}^{(\mathrm{m})}[n], \widetilde{K}^{(\mathrm{s})}, \sigma^2_{w^{(\mathrm{m})}}; \Phi_{\mathrm{C}}\right)$$

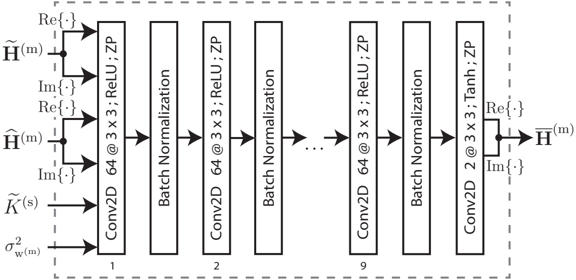


Out-of-Band Aided Channel Estimation - CNN Approach

- input-output mapping relationship with the parameter set of Φ_{C}

$$\overline{\mathbf{H}}^{(\mathrm{m})}[n] = f_{\Phi_{\mathrm{C}}}\left(\widetilde{\mathbf{H}}^{(\mathrm{m})}[n], \widehat{\mathbf{H}}^{(\mathrm{m})}[n], \widetilde{K}^{(\mathrm{s})}, \sigma^2_{w^{(\mathrm{m})}}; \Phi_{\mathrm{C}}\right)$$

- architecture
  - ReLU activation function
  - batch normalization
  - zero padding
  - hyperbolic tangent activation function



Out-of-Band Aided Channel Estimation - UNet Approach

- input-output mapping relationship with the parameter set of Φ_{U}

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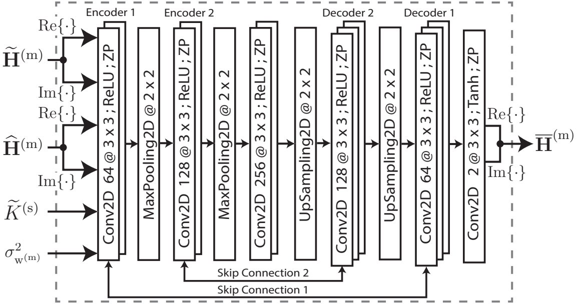

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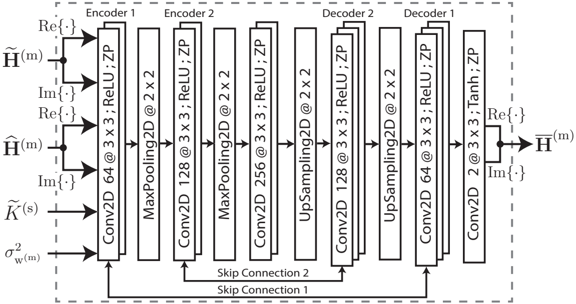
- architecture
  - five convolutional layers + BN layer
  - encoder-decoder structure
  - skip connections
  - hyperbolic tangent activation function



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Simulation-based Comparison

- achievable spectral efficiency (SE)
  - generate dataset
  - simulation parameters

Simulation Parameters

Parameter	**Value**	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Wavelength λ	11.76 cm	1.176 cm
MIMO Configuration	8×8	8×8
Bandwidth B	20.16 MHz	403.2 MHz
Subcarrier Spacing $\triangle f$	60 kHz	120 kHz
Cyclic Prefix t_{CP}	1.19 μs	0.59 μs

Dataset Parameters

Parameter	**Value**
Training Samples L_{train},	500 000
Test Samples L_{test}	500 000
SNR $\gamma^{(\mathrm{m})}$ [dB]	$\mathcal{U} \sim [-20, 10]$
K-factor $K^{(\mathrm{m})}$ [dB]	$\mathcal{U} \sim [-20, 30]$
AoD ϑ, AoA φ [deg]	$\mathcal{U} \sim [-90, 90]$

Simulation-based Comparison

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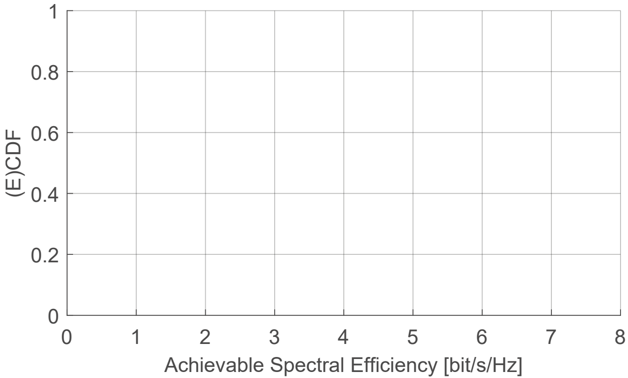
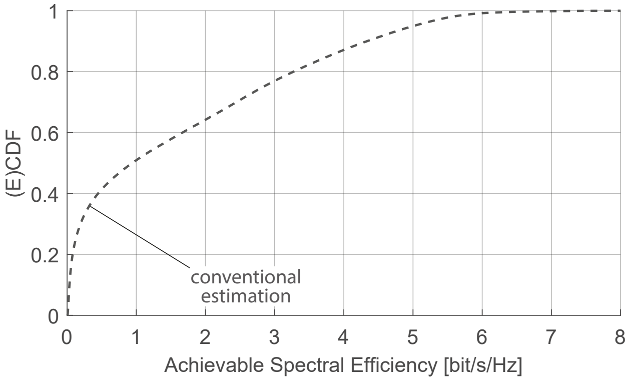


Simulation-based Comparison

- achievable spectral efficiency (SE)
  - generate dataset
  - simulation parameters

- conventional channel estimation



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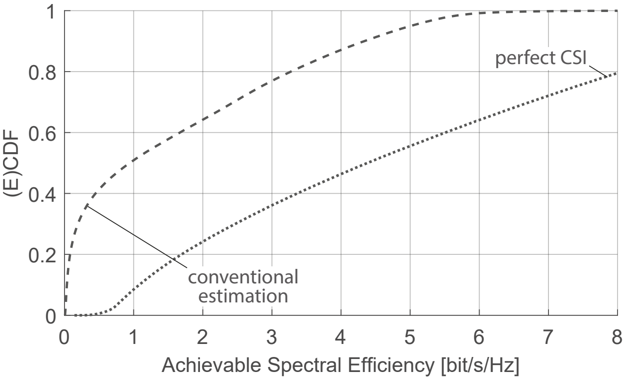
- perfect CSI case



Simulation-based Comparison

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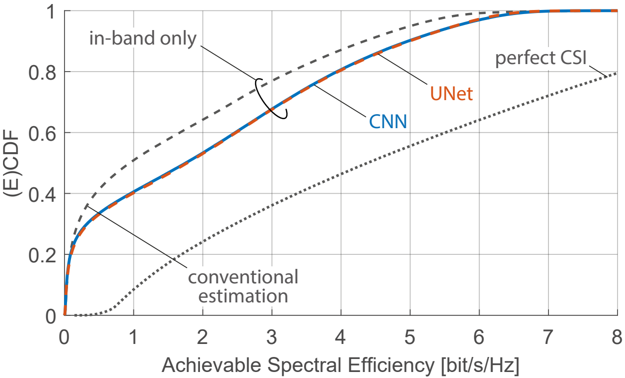
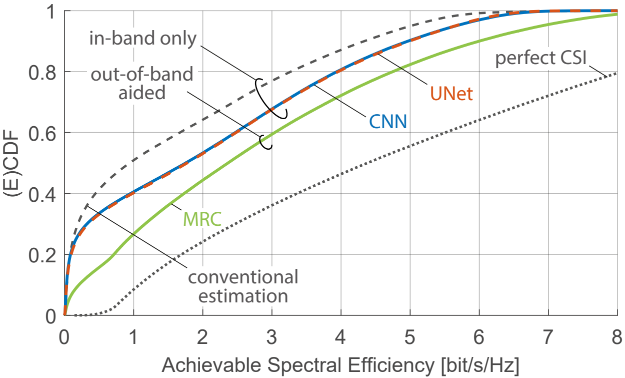
- conventional channel estimation
- perfect CSI case
- in-band CNN/UNet
  - 85% higher median SE



Simulation-based Comparison

- achievable spectral efficiency (SE)
  - generate dataset
  - simulation parameters
- conventional channel estimation
- perfect CSI case
- in-band CNN/UNet
  - 85% higher median SE
- maximal ratio combining (MRC)
  - 150% higher median SE



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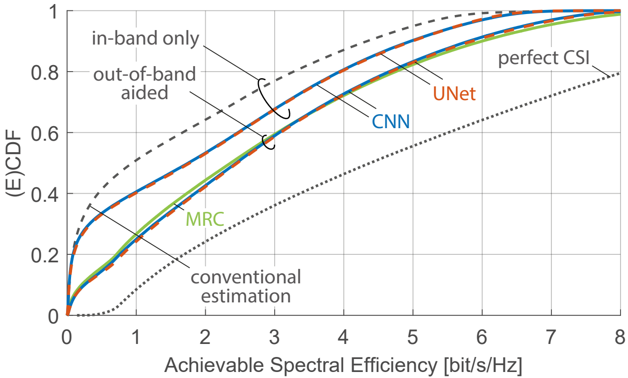
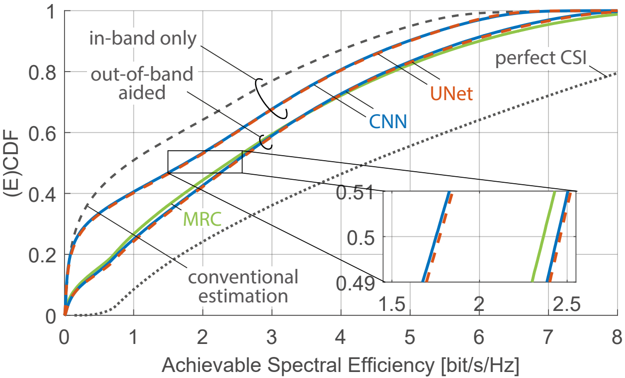
- achievable spectral efficiency (SE)
  - generate dataset
  - simulation parameters
- conventional channel estimation
- perfect CSI case
- in-band CNN/UNet
  - 85% higher median SE
- maximal ratio combining (MRC)
  - 150% higher median SE
- out-of-band aided CNN/UNet
  - 160% higher median SE



Simulation-based Comparison

- achievable spectral efficiency (SE)
  - generate dataset
  - simulation parameters
- conventional channel estimation
- perfect CSI case
- in-band CNN/UNet
  - 85% higher median SE
- maximal ratio combining (MRC)
  - 150% higher median SE
- out-of-band aided CNN/UNet
  - 160% higher median SE



Contents

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Summary and Conclusion

- two novel deep-learning channel estimation methods for mmWave MIMO systems
- CNN and UNet architectures with out-of-band information increase median SE by approximately 3-4%
- UNet architecture demonstrates a slight advantage over the CNN

Thank You !

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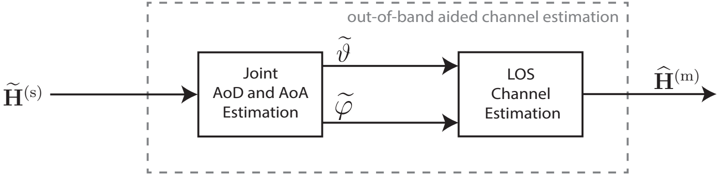
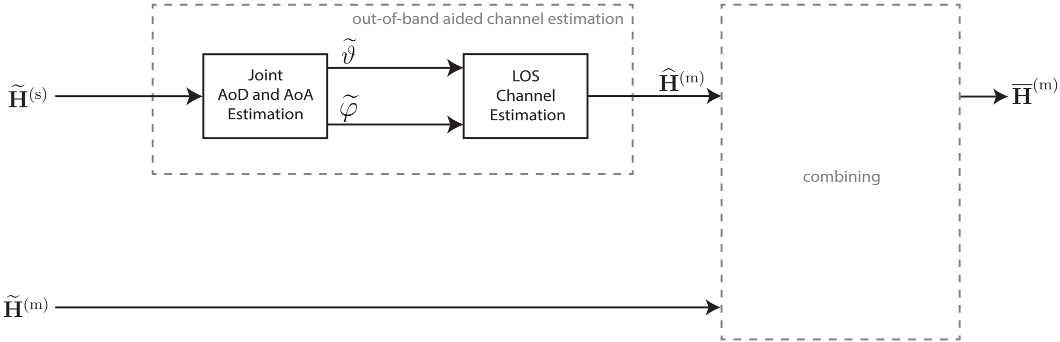
Out-of-Band Aided Channel Estimation



Out-of-Band Aided Channel Estimation



Out-of-Band Aided Channel Estimation - Training/Deployment

- offline training

- online deployment

Out-of-Band Aided Channel Estimation - Training/Deployment

- offline training
  - training set of L_{train} samples
  - considering only the single central subcarrier ($n = N^{(\mathrm{m})}/2$)

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$$\mathrm{NMSE}_{\mathrm{Loss}} = \frac{1}{L_{\text{train}}} \sum_{l=1}^{L_{\text{train}}} \frac{\left\| \mathbf{H}_l^{(\mathrm{m})} - \overline{\mathbf{H}}_l^{(\mathrm{m})} \right\|_F^2}{\left\| \mathbf{H}_l^{(\mathrm{m})} \right\|_F^2}$$

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- online deployment
  - deployment at the receiver for online channel estimation
  - networks are applied for each subcarrier $n \in \{1, \dots, N^{(\mathrm{m})}\}$